

PROJECT EXHIBITION 01

# Breast Cancer Detection Using Histopathology and Machine Learning

## Project Report

A reproducible, explainable CNN-based prototype with an image validation gate and Streamlit interface

| Project Type             | AI/ML educational and research prototype        |
|--------------------------|-------------------------------------------------|
| Primary Task             | Breast histopathology image classification      |
| Output                   | Benign / Malignant / Invalid + model confidence |
| Primary Model            | 4-Stage CNN                                     |
| Evaluation Set           | 1,664 unseen test images; patient-level split   |
| Overall 3-Class Accuracy | 75.96%                                          |

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Dash

## 1. Executive Summary

Project Exhibition 01 is a focused AI/ML prototype for breast histopathology image classification. The system accepts an uploaded image, verifies that the file can be read, preprocesses it, and uses a single convolutional neural network to classify the input as Benign, Malignant, or Invalid. A Streamlit interface presents the prediction and model confidence to the user.

The current evaluation materials report performance on 1,664 test images under a patient-level separation strategy. The computed three-class result is 75.96% accuracy, with 76.19% weighted precision, 75.96% weighted recall, and 76.07% weighted F1-score. Malignant recall is 81.60%, while the invalid class achieves 94.67% recall on 75 invalid samples.

The project remains intentionally compact and reproducible. It does not attempt multimodal cancer prediction, does not expose user-editable thresholds, and keeps benchmark and evaluation reporting separate from the Streamlit user interface.

## 2. Problem Statement

Breast cancer histopathology images contain visual cellular and tissue patterns that can be difficult to interpret consistently and quickly. This project explores how a compact deep-learning pipeline can learn from labelled breast histopathology images and provide an automated preliminary classification while also identifying clearly unsuitable inputs.

The project is not intended to replace a pathologist or provide a clinical diagnosis. Its purpose is to demonstrate the complete AI/ML workflow - from patient-aware data preparation and controlled inference to evaluation and application delivery - in a form suitable for a student exhibition.

## 3. Objectives

- Build a reproducible histopathology image-classification pipeline.
- Validate image files before model inference and support an Invalid output class.
- Train and save a single 4-stage CNN for 3-class prediction.
- Evaluate the final system on a test set separated at patient level.
- Measure overall, class-wise, magnification-level, subtype-level, and invalid-input performance.
- Expose the trained model through a simple Streamlit application.
- Keep the implementation understandable enough for live demonstration and viva explanation.

## 4. Scope and Design Decisions

| Area                | Decision                                                                           |
|---------------------|------------------------------------------------------------------------------------|
| Input               | Breast histopathology image; PNG/JPG/JPEG upload or built-in demo sample           |
| Validation          | Readable-image validation and an Invalid model class for non-histopathology inputs |
| Prediction          | Single CNN; Benign / Malignant / Invalid                                           |
| Input Size          | 128 x 128 RGB                                                                      |
| UI                  | Streamlit                                                                          |
| Backend             | No separate Java/FastAPI service in the current system                             |
| Threshold           | No user-editable malignancy threshold                                              |
| Medical positioning | Educational/research prototype; not a clinical diagnosis                           |

## 5. System Architecture

The final design uses a short inference path. A file is first opened and validated, then converted to RGB, resized, normalized, and passed to the saved 4-stage CNN. The network returns three softmax outputs corresponding to Benign, Invalid, and Malignant. The Streamlit layer displays the selected image, result, confidence, and probability distribution.

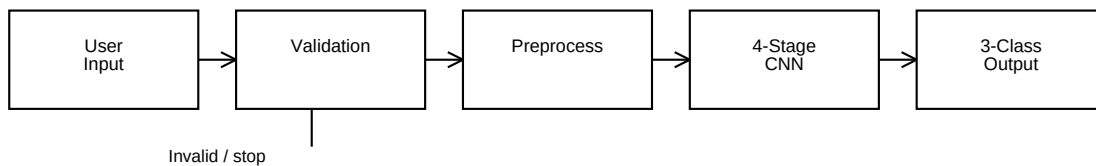


Figure 1. Final end-to-end architecture for Project Exhibition 01.

### 5.1 Main Components

- Streamlit application - handles upload, demo-sample selection, preview, prediction display, probability distribution, and disclaimer.
- Image validation and loading - verifies image integrity, supports common image input types, and converts inputs to RGB.
- Preprocessing - resizes each valid image to 128 x 128 and scales pixel values to [0, 1].
- 4-stage CNN - extracts hierarchical visual features and produces a 3-class softmax output.
- Result layer - displays Benign, Malignant, or Invalid together with model confidence.

## 6. End-to-End Workflow

| Step | Operation                                                                                                                |
|------|--------------------------------------------------------------------------------------------------------------------------|
| 1    | User selects an image using upload mode or one of the built-in demo samples.                                             |
| 2    | The application checks that the image can be opened and verified as a valid image file.                                  |
| 3    | The image is converted to RGB, resized to 128 x 128, normalized to [0, 1], and batched for inference.                    |
| 4    | The saved Keras CNN is loaded from model/breast_cancer_model.keras and used for inference.                               |
| 5    | The network returns a probability vector for Benign, Invalid, and Malignant.                                             |
| 6    | The application displays the predicted class, confidence, detailed probability distribution, and educational disclaimer. |

## 7. Dataset and Data Preparation

The primary dataset used by the project is BreaKHis. The project documentation records 7,909 breast histopathology images from 82 patients, with Benign and Malignant labels and four magnification levels: 40X, 100X, 200X, and 400X. The preparation code groups images by patient identifier extracted from the BreaKHis filenames before creating the splits.

Patient-level separation is a central methodological requirement. Images belonging to the same patient are kept within one split so that related samples are not distributed across training, validation, and test sets. The preparation script uses a 70% / 15% / 15% patient split for training, validation, and testing and then generates separate synthetic non-histopathology images for the Invalid class.

### 7.1 Dataset Preparation Checklist

- Verify Benign and Malignant labels and patient identifiers.
- Group source images by patient before splitting.
- Create patient-level train, validation, and test partitions.
- Generate Invalid-class samples for the three splits using deterministic synthetic patterns.
- Keep training augmentation separate from validation and test preprocessing.
- Keep raw dataset files outside normal source control where appropriate.

## 8. Image Preprocessing

The project uses one consistent preprocessing path so that the representation seen during training matches the representation used during application inference. The documented path is: open image -> verify integrity -> convert to RGB -> resize -> scale pixel values -> add batch dimension.

| Step               | Purpose                                                           |
|--------------------|-------------------------------------------------------------------|
| Image verification | Confirm that the file can be opened and is internally consistent. |
| RGB conversion     | Standardize the channel representation.                           |
| Resize             | Match the 128 x 128 input dimensions expected by the CNN.         |
| Normalization      | Scale pixel values to the [0, 1] range used during training.      |
| Batch expansion    | Produce an input tensor with shape (1, 128, 128, 3).              |

## 9. CNN Model

The current training script defines a lightweight 4-stage CNN with Batch Normalization and Global Average Pooling. It uses increasing feature depth (32, 64, 128, 128), pooling after each convolutional block, dropout between blocks, and a compact classifier head. The final layer is a 3-unit softmax for Benign, Invalid, and Malignant.

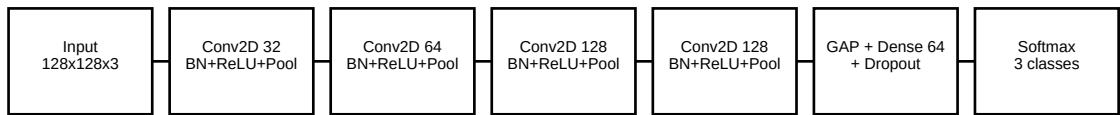


Figure 2. Simplified representation of the current 4-stage CNN architecture.

| Component          | Current configuration                                          |
|--------------------|----------------------------------------------------------------|
| Input              | 128 x 128 x 3 RGB                                              |
| Convolution blocks | 32 -> 64 -> 128 -> 128 filters; 3 x 3 kernels                  |
| Regularization     | Batch Normalization, Dropout, L2 regularization on Dense layer |
| Pooling            | MaxPooling2D after each convolution block                      |
| Classifier head    | Global Average Pooling -> Dense(64, ReLU) -> Dropout(0.30)     |
| Output             | Dense(3, softmax)                                              |
| Optimizer / loss   | Adam (learning rate 0.001) / sparse categorical cross-entropy  |

## 10. Training and Reproducibility

- Create patient-level train, validation, and test partitions.
- Load images at 128 x 128 resolution.
- Apply horizontal and vertical flips, random rotation (0.10), and random zoom (0.10) to training data.
- Normalize training, validation, and test pixels using 1/255 scaling.
- Compute balanced class weights from the training labels.
- Train the CNN for up to 20 epochs with Adam at an initial learning rate of 0.001.
- Use EarlyStopping (patience 5), ReduceLROnPlateau, and ModelCheckpoint on validation loss.
- Save the best model to breast\_cancer\_model.keras and evaluate it on the unseen test split.

## 11. Evaluation Methodology

The evaluation script processes the complete test directory, records predicted class and confidence, extracts magnification and subtype information from BreaKHis filenames, and calculates overall, class-wise, magnification-level, subtype-level, and invalid-input statistics.

| Evaluation view        | Purpose                                                            |
|------------------------|--------------------------------------------------------------------|
| Overall 3-class        | Measures Benign, Invalid, and Malignant behavior together.         |
| Per-class metrics      | Shows precision, recall, and F1 for each class.                    |
| Confusion matrix       | Shows where the model confuses classes.                            |
| Magnification analysis | Checks behavior at 40X, 100X, 200X, and 400X.                      |
| Subtype analysis       | Shows performance across BreaKHis subtypes.                        |
| Invalid-input tests    | Measures non-histopathology rejection and corrupted-file handling. |

## 12. Results

### 12.1 Overall Test Performance

| Metric             | Result |
|--------------------|--------|
| Test samples       | 1,664  |
| Accuracy           | 75.96% |
| Macro Precision    | 80.33% |
| Weighted Precision | 76.19% |
| Macro Recall       | 78.85% |
| Weighted Recall    | 75.96% |
| Macro F1           | 79.56% |
| Weighted F1        | 76.07% |

## 12.2 Per-Class Performance

| Class     | Support | Precision | Recall | F1     |
|-----------|---------|-----------|--------|--------|
| Benign    | 486     | 58.72%    | 60.29% | 59.49% |
| Malignant | 1,103   | 82.27%    | 81.60% | 81.93% |
| Invalid   | 75      | 100.00%   | 94.67% | 97.26% |

The current evaluation uses three output classes. Malignant precision is 82.27% and malignant recall is 81.60%. Benign performance is weaker, with 60.29% recall, while the Invalid class records 94.67% recall. These differences are important because an aggregate accuracy value alone does not show where the model succeeds or fails.

### 12.3 Confusion Matrix

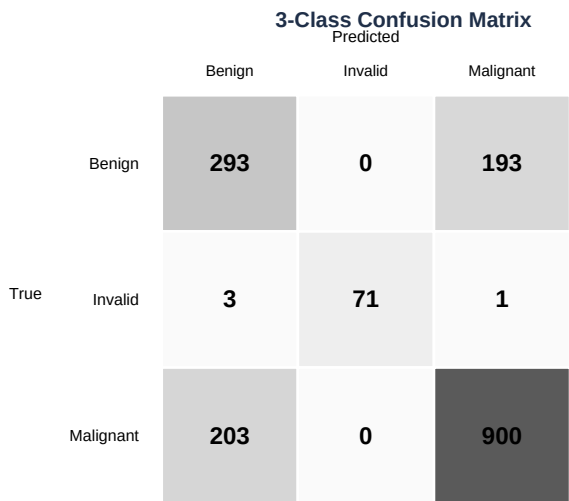


Figure 3. Confusion matrix from the current evaluation report.

The matrix contains 293 correctly predicted Benign images, 71 correctly predicted Invalid images, and 900 correctly predicted Malignant images. The largest cross-class error is 203 true Malignant images predicted as Benign, followed by 193 true Benign images predicted as Malignant.

### 12.4 Magnification-Level Performance

| Magnification | Samples | Accuracy | Weighted Precision | Weighted Recall | Weighted F1 |
|---------------|---------|----------|--------------------|-----------------|-------------|
| 40X           | 392     | 75.00%   | 75.92%             | 75.00%          | 75.37%      |
| 100X          | 422     | 77.73%   | 78.16%             | 77.73%          | 77.91%      |
| 200X          | 389     | 73.01%   | 73.07%             | 73.01%          | 73.04%      |
| 400X          | 386     | 74.35%   | 73.67%             | 74.35%          | 73.92%      |

Magnification performance is reported across the four BreakHis optical zoom levels. The highest accuracy in the current table is 77.73% at 100X, while the lowest is 73.01% at 200X.

## 12.4 Magnification-Level Performance

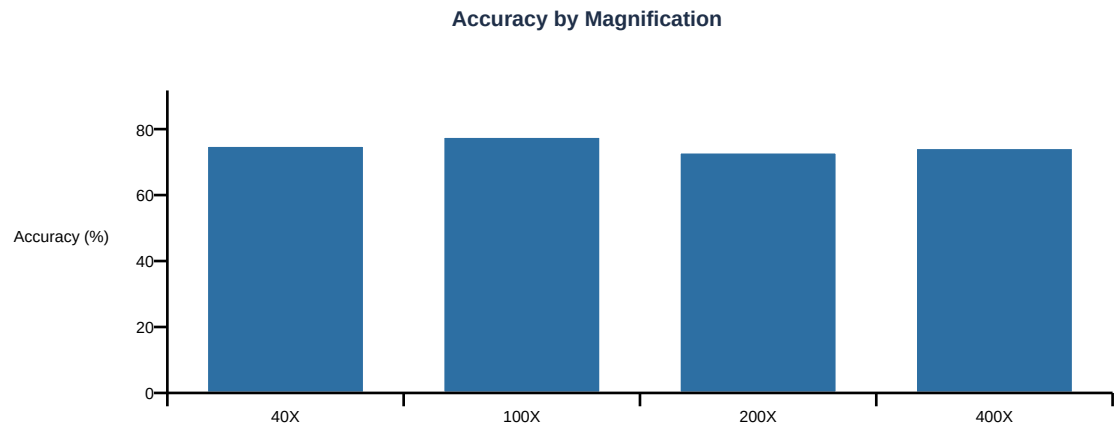


Figure 4. Accuracy by magnification level in the current evaluation.

| Magnification | Samples | Accuracy | Weighted Precision | Weighted Recall | Weighted F1 |
|---------------|---------|----------|--------------------|-----------------|-------------|
| 40X           | 392     | 75.00%   | 75.92%             | 75.00%          | 75.37%      |
| 100X          | 422     | 77.73%   | 78.16%             | 77.73%          | 77.91%      |
| 200X          | 389     | 73.01%   | 73.07%             | 73.01%          | 73.04%      |
| 400X          | 386     | 74.35%   | 73.67%             | 74.35%          | 73.92%      |

The accuracy values range from 73.01% to 77.73% across the four magnifications. The spread is approximately 4.72 percentage points, indicating variation that should be considered when interpreting performance across slide scales.

## 12.5 Histopathological Subtype Performance

| Subtype             | Type      | Samples | Correct | Rate    |
|---------------------|-----------|---------|---------|---------|
| Ductal Carcinoma    | Malignant | 258     | 258     | 100.00% |
| Fibroadenoma        | Benign    | 68      | 0       | 0.00%   |
| Lobular Carcinoma   | Malignant | 125     | 125     | 100.00% |
| Mucinous Carcinoma  | Malignant | 361     | 261     | 72.30%  |
| Papillary Carcinoma | Malignant | 359     | 256     | 71.31%  |
| Phyllodes Tumor     | Benign    | 235     | 131     | 55.74%  |
| Tubular Adenoma     | Benign    | 183     | 162     | 88.52%  |

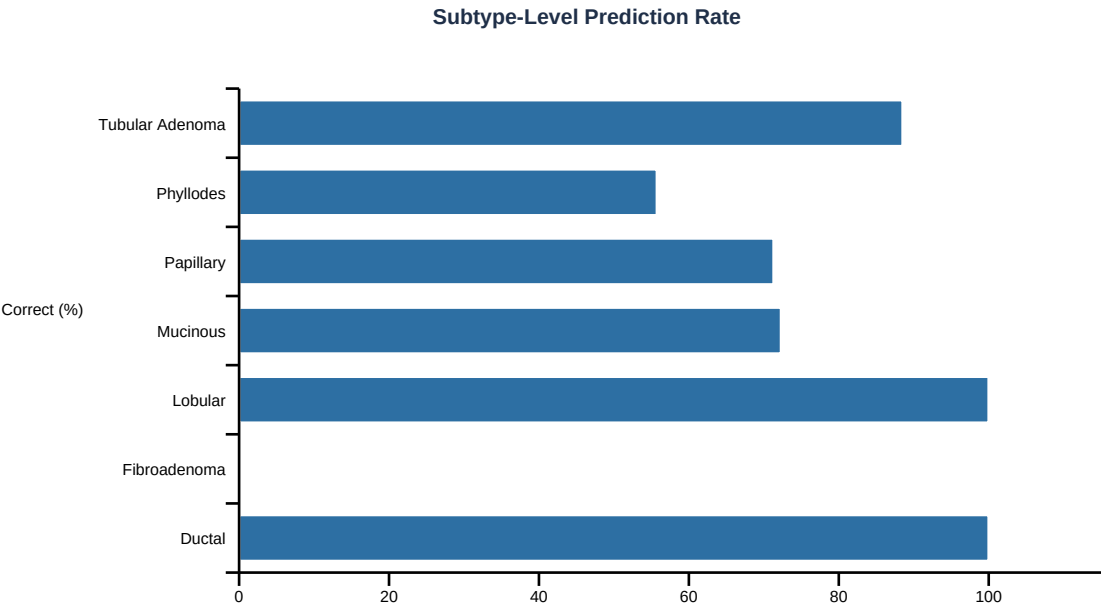


Figure 5. Subtype-level prediction rates from the current evaluation.

Subtype performance is uneven. The model records 100% on Ductal and Lobular Carcinoma in the reported test set, while Fibroadenoma is a major failure case at 0.00%. Phyllodes Tumor also remains substantially weaker at 55.74%. These results show why subtype-level analysis is useful in addition to overall class metrics.

## 12.6 Image Validation Gate

| Gate test                              | Result                              |
|----------------------------------------|-------------------------------------|
| Invalid test samples                   | 75                                  |
| Correctly rejected / predicted Invalid | 71 / 75                             |
| Invalid recall                         | 94.67%                              |
| Invalid precision                      | 100.00%                             |
| Corrupted / invalid file handling      | Passed in supplied evaluation tests |

Input Validation Test - 75 Invalid Samples

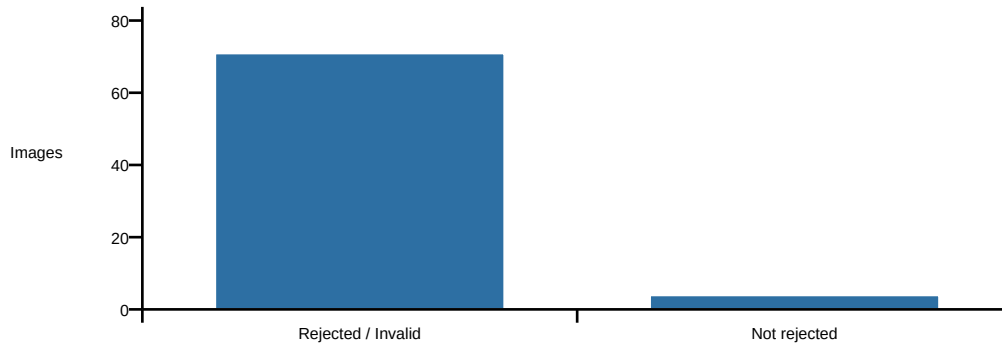


Figure 6. Input-gate performance on the current invalid-image test set.

The evaluation shows that 71 of 75 invalid test samples were assigned to the Invalid class. Corrupted input is handled separately in the prediction code by trapping image-loading and verification errors before CNN inference.

## 13. Evaluation Quality Note

The current MODEL\_EVALUATION\_REPORT.md contains internally inconsistent narrative summary lines. Its computed metric table and confusion matrix support 75.96% three-class accuracy, 82.27% malignant precision, 81.60% malignant recall, and 94.67% invalid recall. Later hard-coded narrative lines state 74.10% accuracy and different malignant and invalid figures. This revised report uses the computed table and confusion matrix because they reconcile arithmetically and are the safer basis for documentation.

## 14. Application Design

The application layer is deliberately small. Streamlit provides the interface without a separate web backend. The current interface supports both direct upload and a built-in demo-sample mode.

| Screen / Step            | Current behavior                                                                                         |
|--------------------------|----------------------------------------------------------------------------------------------------------|
| Input mode               | Choose between uploading an image and using one of five built-in demo samples.                           |
| Preview                  | Selected image is shown before prediction.                                                               |
| Validation / inference   | Readable images are processed through the same prediction function; missing/corrupt inputs are reported. |
| Prediction               | Result is shown as Benign, Malignant, or Invalid.                                                        |
| Confidence               | Top model output is displayed as a percentage with a progress indicator.                                 |
| Probability distribution | Expandable section shows the three class probabilities.                                                  |
| Safety notice            | Educational and research disclaimer remains visible.                                                     |

## 15. Testing Strategy

- Model-file existence and load test.
- Benign sample inference test.
- Malignant sample inference test.
- Invalid non-histopathology sample test.
- Corrupted-file handling test.
- Built-in demo sample validation for benign, malignant, and invalid examples.
- Result display and confidence-bound checks.

## 16. Limitations

- Performance depends strongly on the quality and diversity of the training data.
- The current model shows weaker benign-class performance than malignant performance.
- Subtype behavior is uneven, especially for Fibroadenoma and Phyllodes Tumor in the reported test set.
- The Invalid class is based on synthetic non-histopathology images for training and evaluation.
- The model may encounter staining, magnification, and acquisition variations not fully represented in the training data.
- Model confidence is an output score and should not be interpreted as a clinically calibrated probability.
- The system is an educational/research prototype and must not be used as a clinical diagnostic system.

## 17. Future Improvements

- Improve the Invalid class using a larger and more representative negative-image set.
- Investigate benign-subtype confusion, particularly Fibroadenoma and Phyllodes Tumor.
- Evaluate additional preprocessing and augmentation strategies based on observed failure modes.
- Compare carefully chosen CNN or transfer-learning architectures while maintaining the same patient-level test protocol.
- Add explainability methods such as Grad-CAM for demonstration and error analysis.
- Automate metric summaries so report text is generated from the same computed results.
- Consider cloud deployment only after local reproducibility and evaluation are stable.

## 18. Reproducibility and Repository Structure

The current repository keeps the runnable application, saved model, sample inputs, and training/evaluation scripts together. The key files are:

| Path                            | Purpose                                                       |
|---------------------------------|---------------------------------------------------------------|
| app.py                          | Streamlit application interface                               |
| predict.py                      | Image validation, preprocessing, model loading, and inference |
| model/breast_cancer_model.keras | Saved Keras model used by the application                     |
| sample_images/                  | Built-in demonstration inputs                                 |
| train_model.py                  | 4-stage CNN training and checkpointing pipeline               |
| prepare_data.py                 | Patient-level splitting and Invalid-class generation          |
| evaluate_detailed.py            | Detailed test-set evaluation and analysis                     |
| test_system.py                  | Automated system verification tests                           |
| requirements.txt                | Python package dependencies                                   |
| MODEL_EVALUATION_REPORT.md      | Current evaluation snapshot                                   |

## 19. Responsible Use and Medical Positioning

This system is an AI/ML educational and research prototype. It is not a medical device and does not replace a qualified medical professional. Prediction confidence represents model output and should not be interpreted as a guaranteed medical probability.

## 20. Team and Contributions

| Role                  | Name                 |
|-----------------------|----------------------|
| Author / Project Lead | Shwetank Vaibhav     |
| Contributor           | Pranshu Gupta        |
| Contributor           | Ayush Shukla         |
| Contributor           | Debasish Kumar Sahoo |
| Contributor           | Manvendra Kumar      |
| Contributor           | Piyush Kumar Dash    |

## 21. Conclusion

Project Exhibition 01 demonstrates a complete, compact AI/ML pipeline for breast histopathology image classification, combining patient-level data preparation, controlled image validation, unified preprocessing, a 4-stage CNN, quantitative evaluation, and a Streamlit application.

The current evaluation reports 75.96% accuracy across 1,664 test images, with 81.60% malignant recall and 94.67% Invalid-class recall. The results also reveal weaker benign classification and selected benign subtypes, which remain important limitations.

Overall, the project provides a reproducible student-focused demonstration of medical image classification and application deployment. Future refinement should be driven by observed failure cases and stronger evaluation discipline.

### Appendix A. Key Commands

```
python -m venv venv
pip install -r requirements.txt
streamlit run app.py
```

### Appendix B. Primary Evaluation Snapshot

Test set: 1,664 images | Accuracy: 75.96% | Weighted precision: 76.19% | Weighted recall: 75.96% | Weighted F1: 76.07% | Invalid recall: 94.67%